NCRB Data Analysis

Problem Statement

The objective of this analysis is to study and visualize crime-related data across different states and regions. The patterns are:

- Number of apprehended cases
- Acquitted cases
- Pending cases
- Crime status distribution

Using statistical charts, the goal is to identify trends, relationships, and regional variations in crime handling.

Data Source

The dataset used for this analysis is based on records inspired by the National Crime Records Bureau (NCRB), Government of India. Get this from *"data.gov.in"*

It contains the following attributes:

- **State**
- **Region**
- **Year**
- **Crime Status (High / Low)**
- **Apprehended Cases**
- **Acquitted Cases**
- **Pending Cases**

Pre Processing:

```r
install.packages(c("readxl", "vcd"))
```

```
## Installing packages into '/cloud/lib/x86_64-pc-linux-gnu-library/4.5'
## (as 'lib' is unspecified)
```

```r
library(readxl)
library(vcd)
```

```
## Loading required package: grid
```

```r
data <- read_excel("NCRB dataset.xlsx")
str(data)
```

```
## tibble [36 × 7] (S3: tbl_df/tbl/data.frame)
## $ state        : chr [1:36] "Andhra Pradesh" "Arunachal Pradesh" "Assam" "
Bihar" ...
## $ region       : chr [1:36] "South" "North-East" "North-East" "East" ...
##  $ crime_status: chr [1:36] "Low" "Low" "Low" "High" ...
##  $ year        : num [1:36] 2023 2023 2023 2023 2023 ...
##  $ apprehended : num [1:36] 901 34 146 2457 2161 ...
##  $ acquitted   : num [1:36] 197 6 23 163 211 10 72 359 39 54 ...
##  $ pending     : num [1:36] 997 76 289 2393 2923 ...
```

```r
head(data)
```

```
## # A tibble: 6 × 7
##   state             region     crime_status  year apprehended acquitted pe
nding
##   <chr>             <chr>      <chr>        <dbl>       <dbl>     <dbl>
<dbl>
## 1 Andhra Pradesh    South      Low           2023         901       197
997
## 2 Arunachal Pradesh North-East Low           2023          34         6
76
## 3 Assam             North-East Low           2023         146        23
289
## 4 Bihar             East       High          2023        2457       163
2393
## 5 Chhattisgarh      Central    High          2023        2161       211
2923
## 6 Goa               West       Low           2023          43        10
98
```

Data Subsetting

```
high_crime

## # A tibble: 11 × 7
##    state          region  crime_status  year apprehended acquitted pending
##    <chr>          <chr>   <chr>        <dbl>       <dbl>     <dbl>   <dbl>
##    Bihar          East    High          2023        2457       163    2393
##    Chhattisgarh   Central High          2023        2161       211    2923
##    Gujarat        West    High          2023        1999        72    1733
##    Haryana        North   High          2023        1986       359    1772
##    Madhya Pradesh Central High          2023        4330       598    5695
##    Maharashtra    West    High          2023        5280       466    8976
##    Rajasthan      West    High          2023        3898        79    1785
##    Tamil Nadu     South   High          2023        3811       917    4906
##    Uttar Pradesh  Central High          2023        1822        97    1943
## 10 West Bengal    East    High          2023         925       145    1175
## 11 Delhi          North   High          2023        3098       147    1894

south_region

## # A tibble: 6 × 7
##   state          region crime_status  year apprehended acquitted pending
##   <chr>          <chr>  <chr>        <dbl>       <dbl>     <dbl>   <dbl>
## 1 Andhra Pradesh South  Low           2023         901       197     997
## 2 Karnataka      South  Low           2023        1072       105     525
## 3 Kerala         South  Low           2023         789        29     370
## 4 Tamil Nadu     South  High          2023        3811       917    4906
## 5 Telangana      South  Low           2023        1467       196     890
## 6 Puducherry     South  Low           2023         144         8     579

high_apprehended

## # A tibble: 12 × 7
##    state          region  crime_status  year apprehended acquitted pending
##    <chr>          <chr>   <chr>        <dbl>       <dbl>     <dbl>   <dbl>
##    Bihar          East    High          2023        2457       163    2393
##    Chhattisgarh   Central High          2023        2161       211    2923
##    Gujarat        West    High          2023        1999        72    1733
##    Haryana        North   High          2023        1986       359    1772
##    Madhya Pradesh Central High          2023        4330       598    5695
##    Maharashtra    West    High          2023        5280       466    8976
##    Odisha         East    Low           2023        1507         0      59
##    Rajasthan      West    High          2023        3898        79    1785
##    Tamil Nadu     South   High          2023        3811       917    4906
## 10 Telangana      South   Low           2023        1467       196     890
## 11 Uttar Pradesh  Central High          2023        1822        97    1943
## 12 Delhi          North   High          2023        3098       147    1894

backlog_states
```

```
## # A tibble: 19 × 7
##    state                  region crime_status  year apprehended acquitted
pending
##    <chr>                  <chr>  <chr>         <dbl>       <dbl>     <dbl>
<dbl>
##  1 Andhra Pradesh         South  Low            2023         901       197
997
##  2 Arunachal Pradesh      North… Low            2023          34         6
76
##  3 Assam                  North… Low            2023         146        23
289
##  4 Chhattisgarh           Centr… High           2023        2161       211
2923
##  5 Goa                    West   Low            2023          43        10
98
##  6 Himachal Pradesh       North  Low            2023         244        39
275
##  7 Jharkhand              East   Low            2023         190        54
229
##  8 Madhya Pradesh         Centr… High           2023        4330       598
5695
##  9 Maharashtra            West   High           2023        5280       466
8976
## 10 Manipur                North… Low            2023          25         0
64
## 11 Meghalaya              North… Low            2023          88        31
198
## 12 Mizoram                North… Low            2023          19         0
41
## 13 Sikkim                 North… Low            2023          15         1
25
## 14 Tamil Nadu             South  High           2023        3811       917
4906
## 15 Uttar Pradesh          Centr… High           2023        1822        97
1943
## 16 West Bengal            East   High           2023         925       145
1175
## 17 Andaman and Nicobar …  Island Low            2023          25        10
150
## 18 Jammu and Kashmir      North  Low            2023         420       119
664
## 19 Puducherry             South  Low            2023         144         8
579
```

Central Tendency Measures

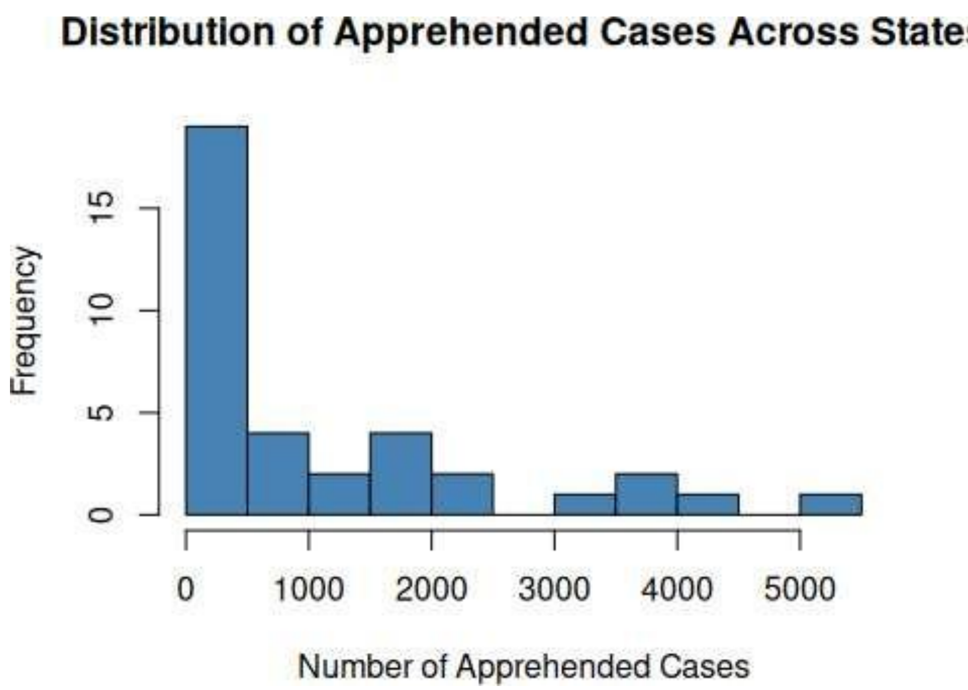
```r
mean(data$apprehended)

## [1] 1112.111
```

```r
median(data$apprehended)

## [1] 332

mean(data$acquitted)

## [1] 118.6111

median(data$acquitted)

## [1] 46.5

mean(data$pending)

## [1] 1155.111

median(data$pending)

## [1] 329.5
```

Histogram Analysis

Distribution of Apprehended Cases Across States

```
hist(
  data$apprehended,
  breaks = 10,
  col = "steelblue",
  border = "black",
  main = "Distribution of Apprehended Cases Across States",
  xlab = "Number of Apprehended Cases",
  ylab = "Frequency"
)
```



Insights

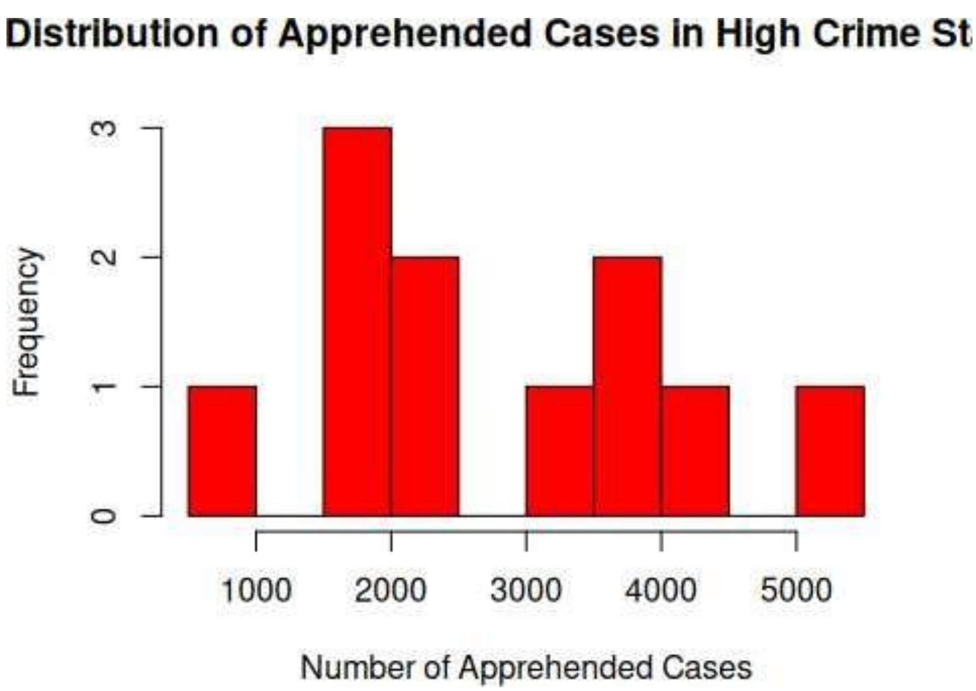
• The average number of apprehended cases is 1112, which is much higher than the median of 332. This suggests a skewed distribution, where a few states have very high arrest numbers.
• Most states fall below the average, indicating that lower apprehension counts are more common in the dataset.
• Only a few states report extremely high apprehended cases, resulting in a long right tail.

Inference

• Crime enforcement varies significantly across states. A few states manage a much higher number of arrests.
• Policymakers may need to explore whether higher crime rates or more effective policing causes these high numbers.

Distribution of Apprehended Cases in High Crime States

```r
hist(
  high_crime$apprehended,
  breaks = 10,
  col = "red",
  border = "black",
  main = "Distribution of Apprehended Cases in High Crime States",
  xlab = "Number of Apprehended Cases",
  ylab = "Frequency"
)
```



Insights

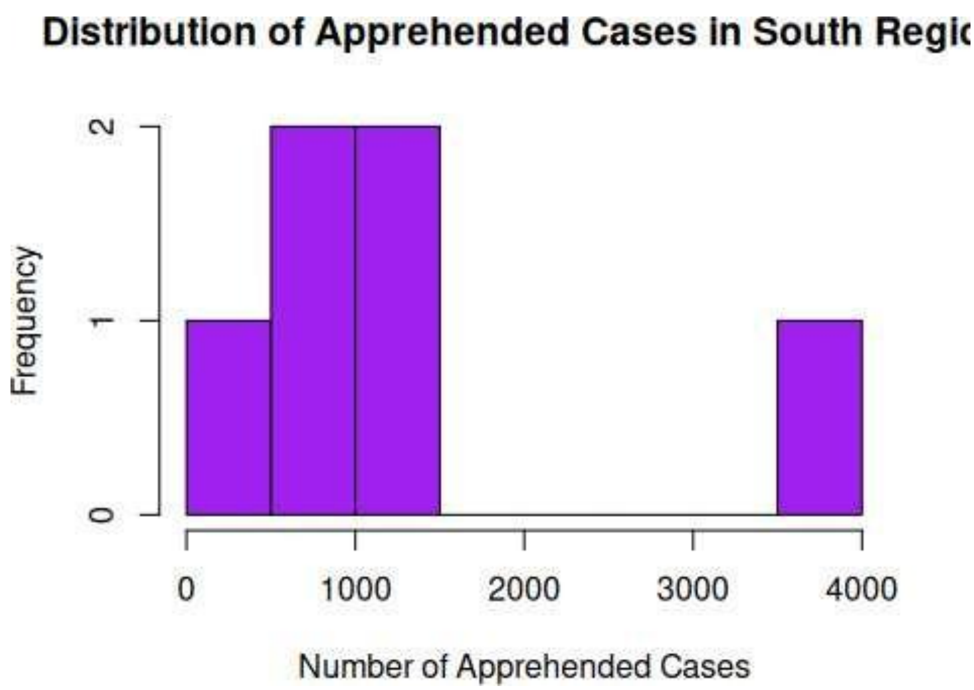
• High crime states have higher apprehended values grouped in larger ranges compared to the entire dataset.
• The distribution is still right skewed, meaning that a few states report very large arrest numbers.
• The data spread is wider, showing significant differences among high crime states.

Inference

• Even within high crime states, law enforcement results vary a lot, which suggests differences in policing effectiveness or crime levels.

Distribution of Apprehended Cases in South Region

```r
hist(
  south_region$apprehended,
  breaks = 10,
  col = "purple",
  border = "black",
  main = "Distribution of Apprehended Cases in South Region",
  xlab = "Number of Apprehended Cases",
  ylab = "Frequency"
)
```



Insights

• Compared to states with high crime rates, the South region exhibits moderate apprehended values with fewer extreme outliers.
• Rather than the mean, the majority of observations fall closer to the median range.
• There is still a small right skewness in the distribution.

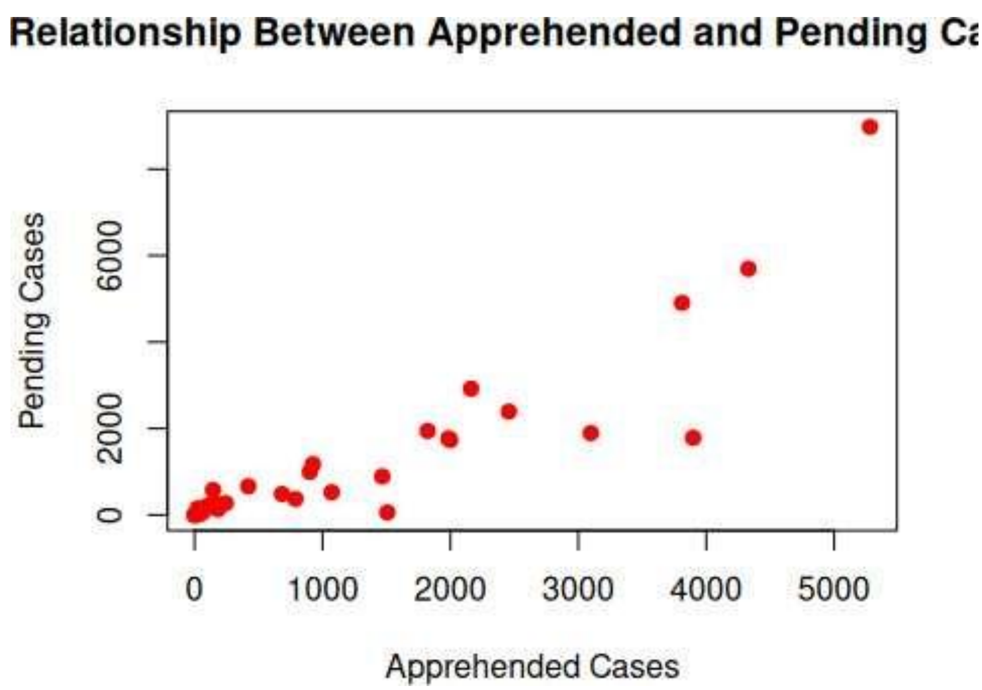
Inference

• The South seems to have more balanced crime control, with fewer states having exceptionally high arrest rates.

Scatter Plot Analysis

Relationship Between Apprehended and Pending Cases

```
plot(
  data$apprehended,
  data$pending,
  col = "red",
  pch = 19,
  main = "Relationship Between Apprehended and Pending Cases",
  xlab = "Apprehended Cases",
  ylab = "Pending Cases"
)
```



Insights

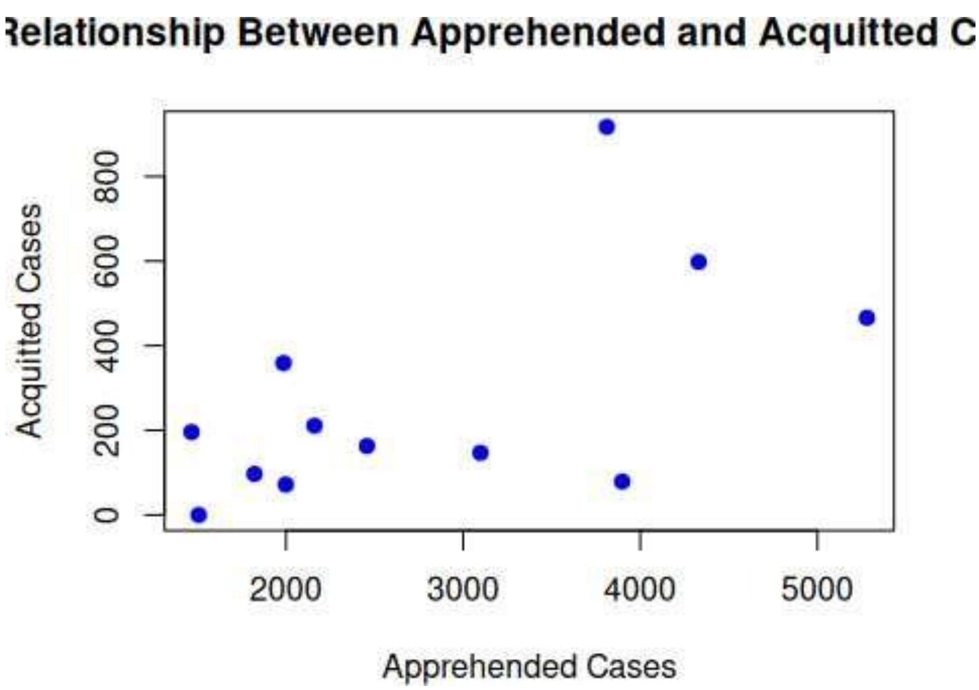
• The scatter plot demonstrates a favorable correlation between cases that have been apprehended and those that are still pending.
• There are typically more pending cases in states with a higher number of apprehended cases.• This implies that legal or judicial procedures might not keep up with the rate of arrests.

Inference

• The backlog of cases grows with the volume of crimes; increased arrest activity does not always translate into quicker case resolution.

Relationship Between Apprehended and Acquitted Cases

```
plot(
  high_apprehended$apprehended,
  high_apprehended$acquitted,
  col = "blue",
  pch = 19,
  main = "Relationship Between Apprehended and Acquitted Cases",
  xlab = "Apprehended Cases",
  ylab = "Acquitted Cases"
)
```



Insights

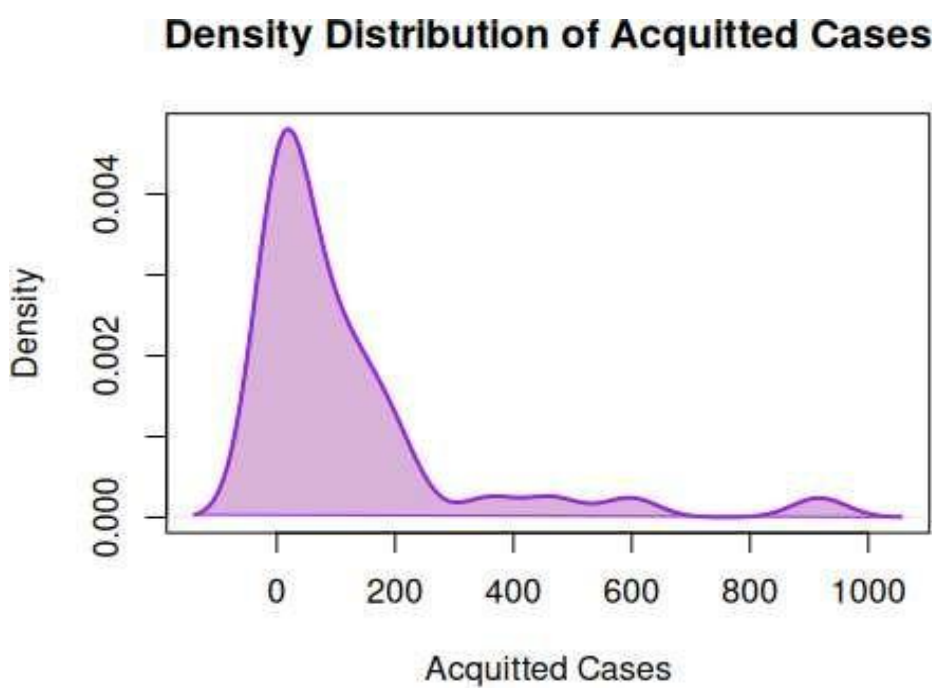
• The correlation between cases that are apprehended and those that are acquitted is somewhat positive.
• States with higher rates of arrests also typically have higher rates of acquittals.
• Variability in conviction outcomes is indicated by the spread of points.

Inference

• A rise in arrests could result in a higher number of acquittals because of shaky evidence or court hold-ups, indicating potential problems with case processing.

Density Plot

```
d <- density(data$acquitted)

plot(
  d,
  col = "purple",
  lwd = 2,
  main = "Density Distribution of Acquitted Cases",
  xlab = "Acquitted Cases",
  ylab = "Density"
)

polygon(d, col = rgb(0.5,0,0.5,0.3), border = "purple")
```



Insights

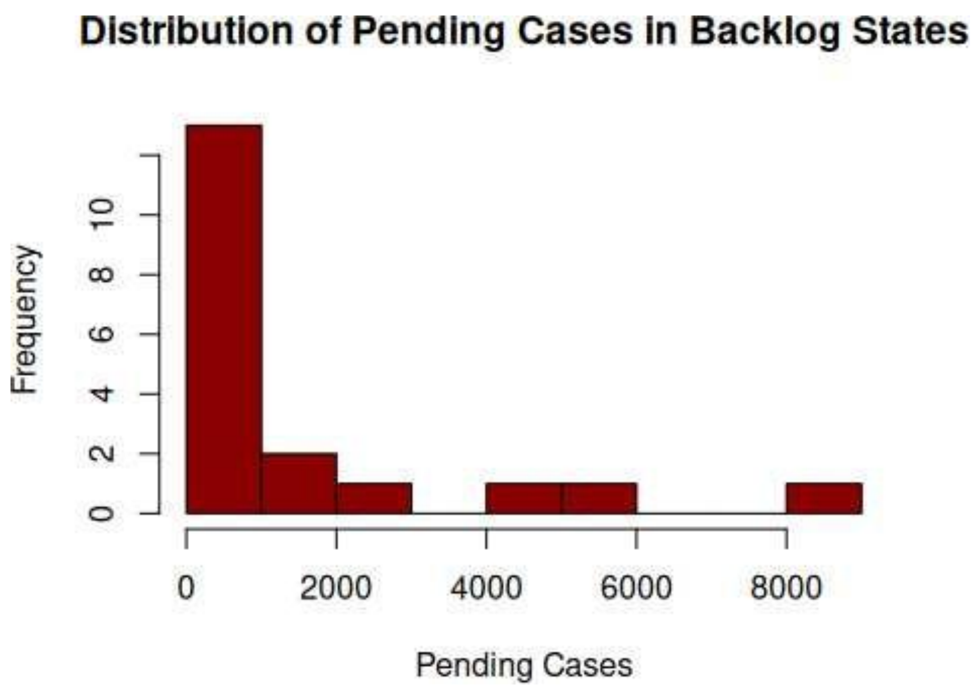
• The density curve peaks at lower values, indicating that the number of acquittals in the majority of states is comparatively low.
• The distribution is positively skewed because the mean is greater than the median.
• The distribution is stretched by a few states with extremely high acquittal rates.

Inference

• Acquittal results differ greatly between states, indicating variations in the caliber of investigations, the strength of the evidence, or the effectiveness of the legal system.

Pending Cases Distribution

```
hist(
  backlog_states$pending,
  breaks = 10,
  col = "darkred",
  border = "black",
  main = "Distribution of Pending Cases in Backlog States",
  xlab = "Pending Cases",
  ylab = "Frequency"
)
```



Insights

• A significant right skewness in pending cases is indicated by the wide difference between the mean and median.
• While the majority of states have comparatively small backlogs of cases, a few report exceptionally high numbers.
• This suggests that different regions handle cases differently.

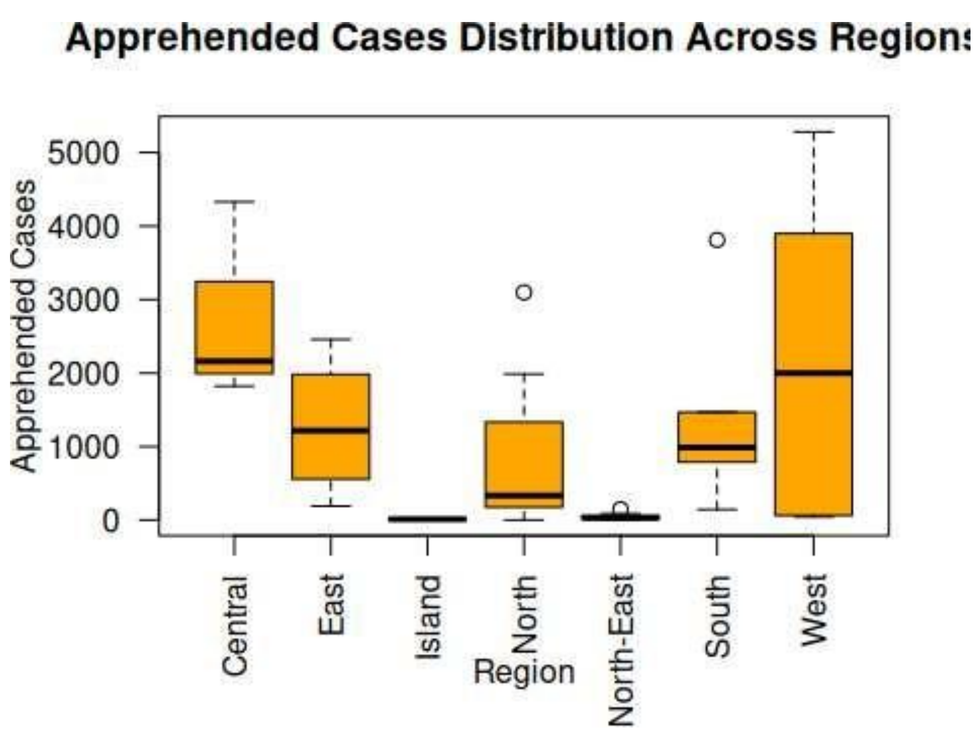
Inference

• There may be a significant backlog of cases in some states, necessitating better case management and quicker legal processing.

Boxplot Analysis

Apprehended Cases Distribution Across Regions

```r
boxplot(
  apprehended ~ region,
  data = data,
  col = "orange",
  main = "Apprehended Cases Distribution Across Regions",
  xlab = "Region",
  ylab = "Apprehended Cases",
  las = 2
)
```



Insights

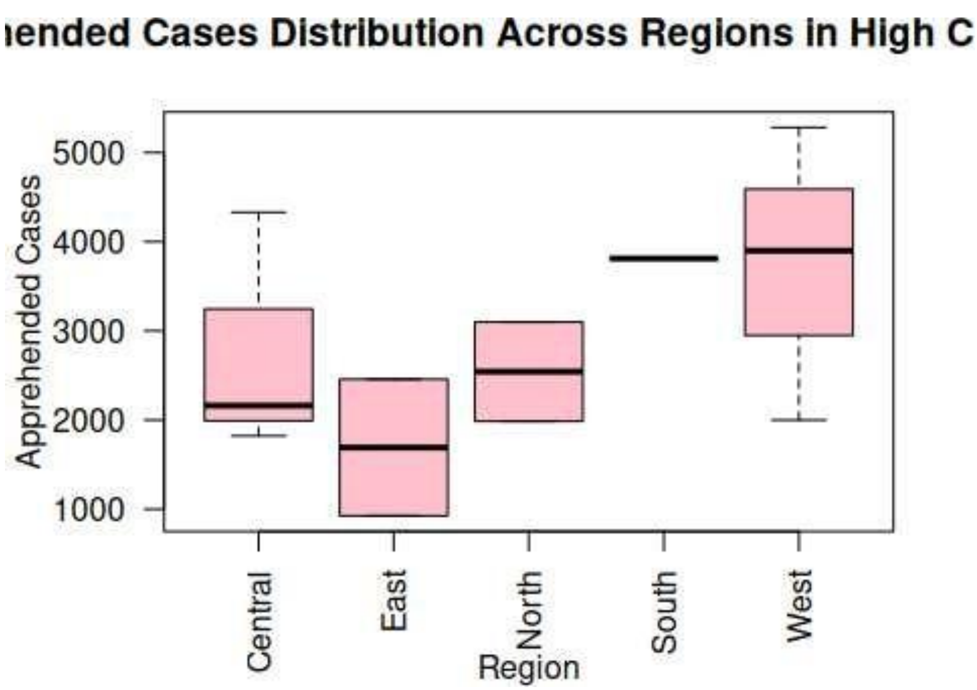
- The boxplot shows high variability in apprehended cases among regions.
- Some regions display extreme values and outliers above the upper quartile.
- The median line differs between regions, indicating different crime enforcement levels.

Inference

- Crime enforcement patterns vary significantly across regions due to population size, policing resources, or crime rates.

Apprehended Cases Distribution in High Crime States

```
boxplot(
  apprehended ~ region,
  data = high_crime,
  col = "pink",
  main = "Apprehended Cases Distribution Across Regions in High Crime
States",
  xlab = "Region",
  ylab = "Apprehended Cases",
  las = 2
)
```



Insights

• States with high rates of crime have higher median apprehended values than the overall dataset.
• There is more variation among high-crime areas, as indicated by the wider interquartile range.• Outliers indicate that the number of arrests in some states is abnormally high.

Inference

• Because crime enforcement and case loads in high-crime states differ significantly from those in other areas, targeted policy interventions are necessary.

DASHBOARD:

Central Tendency Code

```
mean(data$apprehended)
median(data$apprehended)

mean(data$acquitted)
median(data$acquitted)

mean(data$pending)
median(data$pending)
```

Central Tendency Output

```
## Mean of Apprehended Cases: 1112.111
```

```
## Median of Apprehended Cases: 332
```

```
## Mean of Acquitted Cases: 118.6111
```

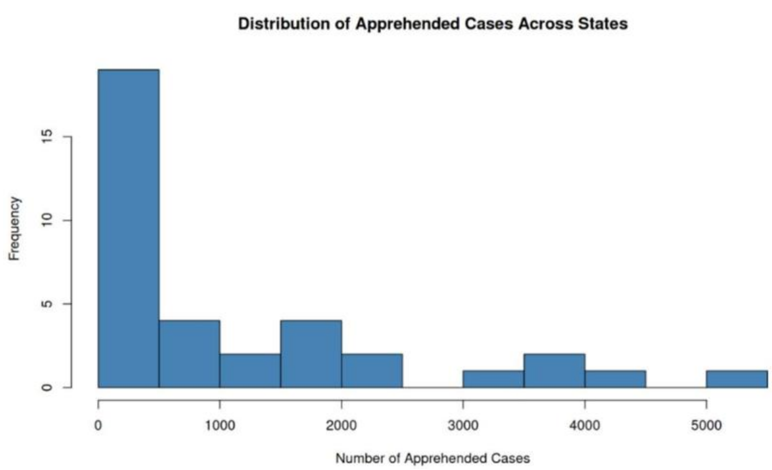
```
## Median of Acquitted Cases: 46.5
```

```
## Mean of Pending Cases: 1155.111
```

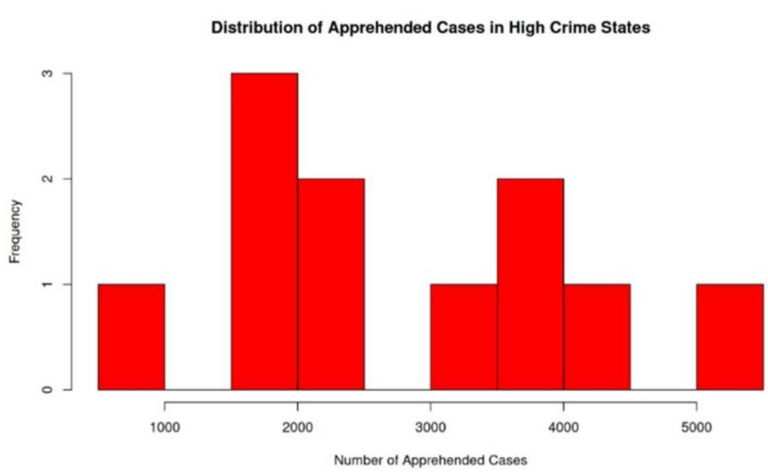
```
## Median of Pending Cases: 329.5
```

Histogram Analysis:

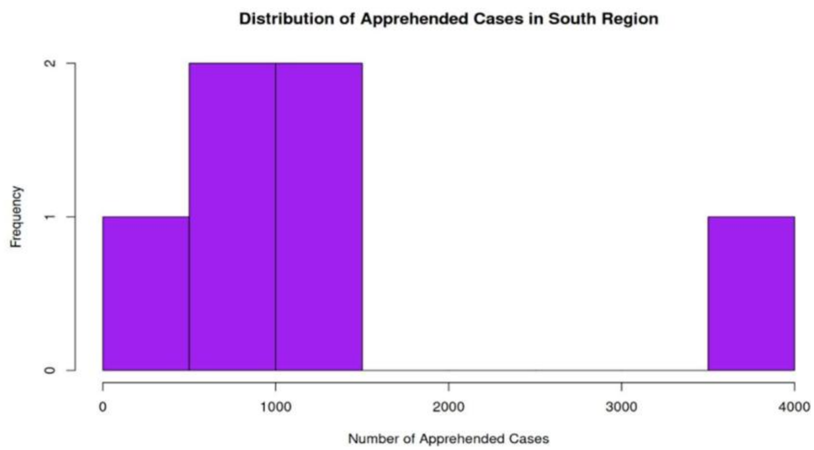
Distribution of Apprehended Cases Across States



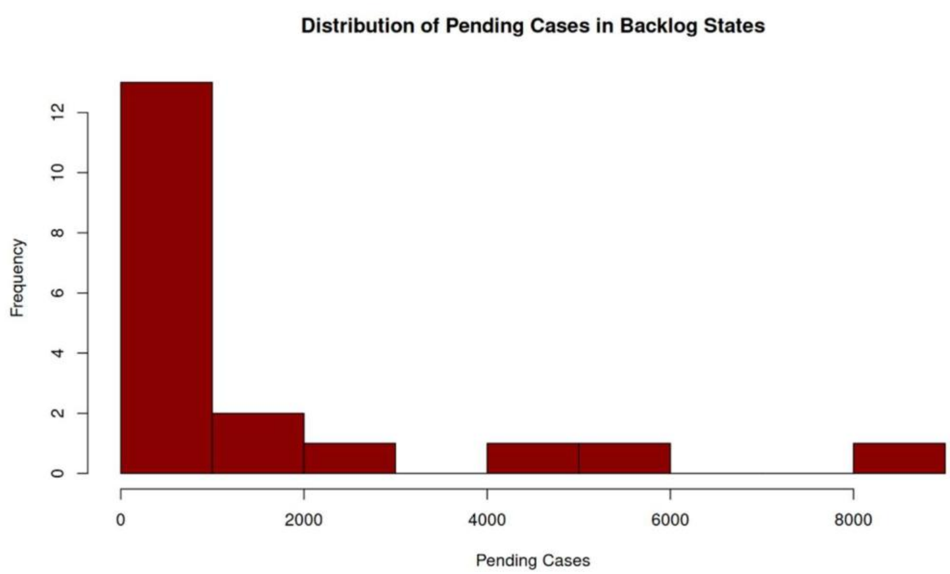
Distribution of Apprehended Cases in High Crime States



Distribution of Apprehended Cases in South Region

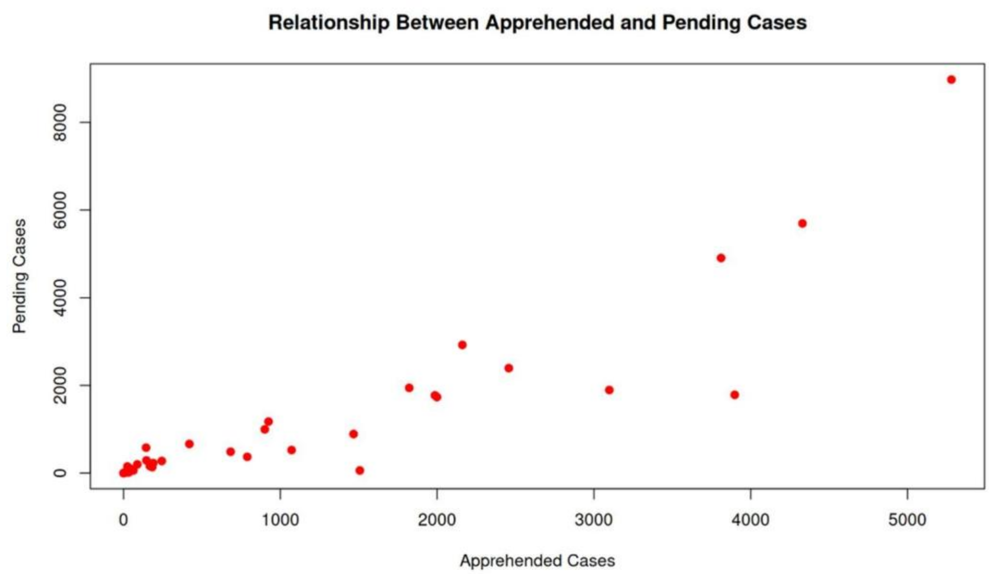


Distribution of Pending Cases in Backlog States

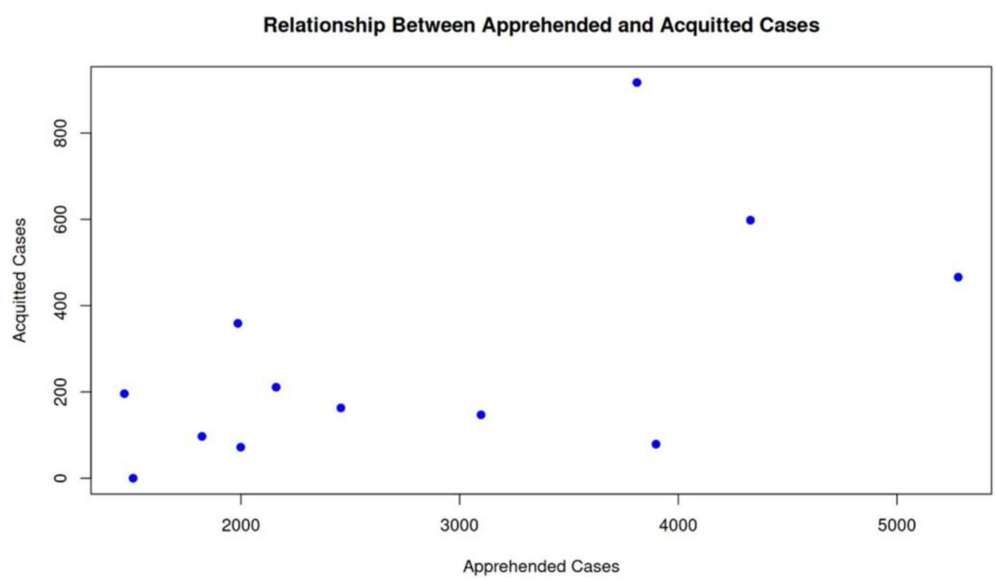


Scatter plot Analysis:

Relationship Between Apprehended and Pending Cases

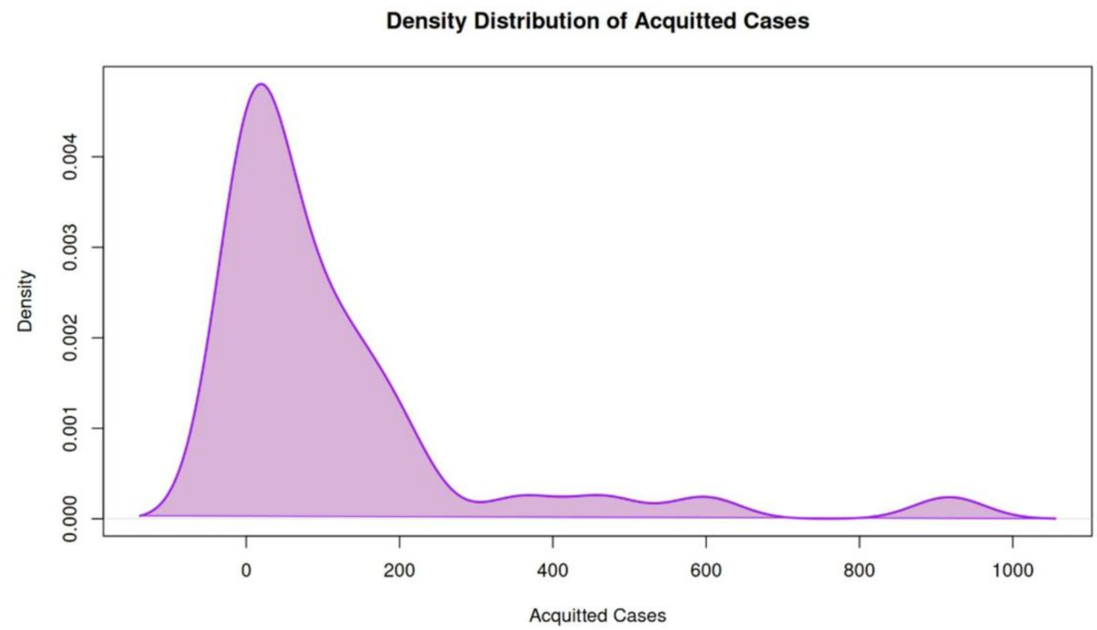


Relationship Between Apprehended and Acquitted Cases



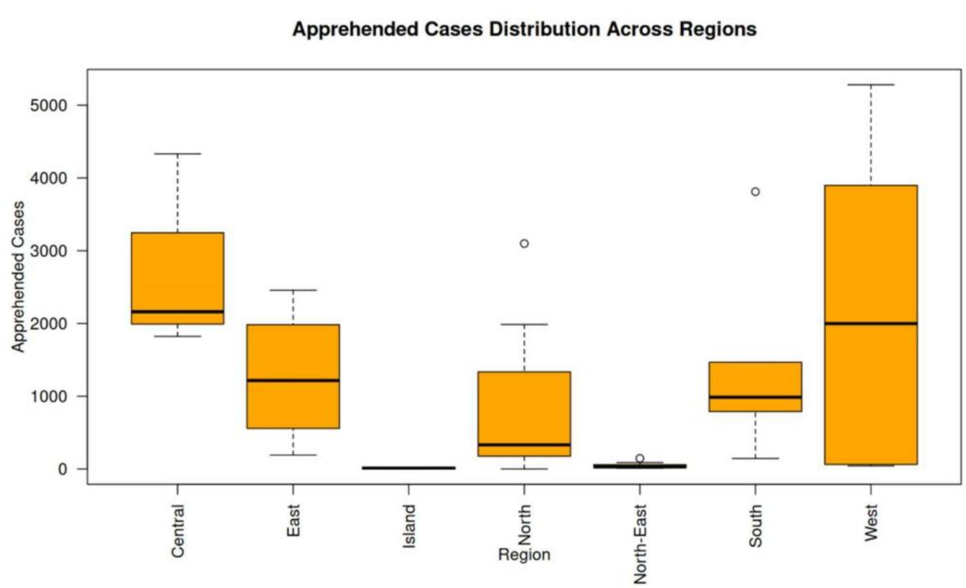
Density Plot Analysis:

Density Distribution of Acquitted Cases

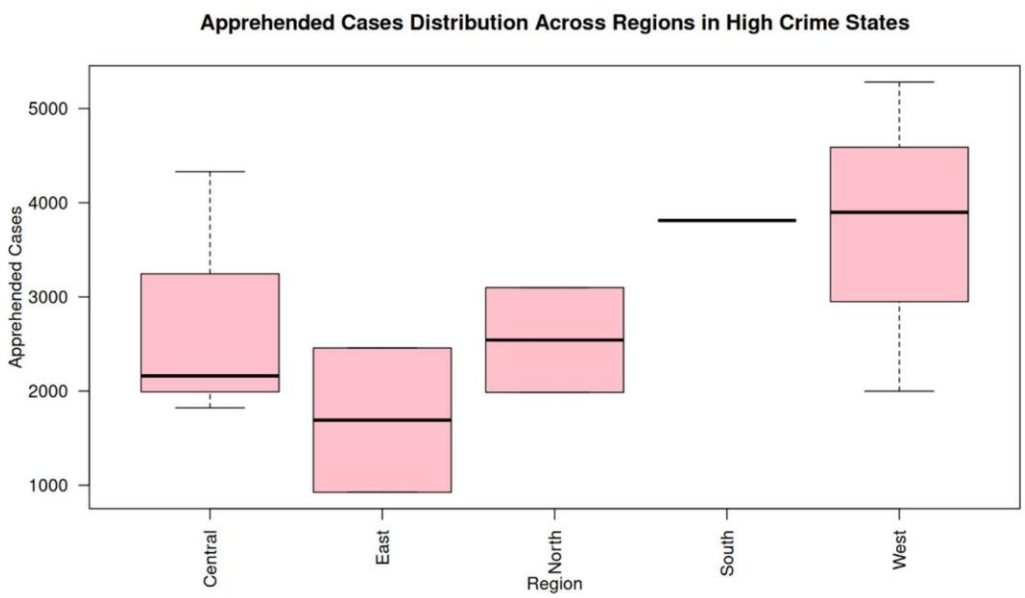


Boxplot Analysis

Apprehended Cases Distribution Across Regions



Apprehended Cases Distribution Across Regions in High Crime States



Story Summary:

The NCRB crime data set includes information on apprehended cases, acquitted cases, pending cases, crime status, and regional distribution across different states. In the study, descriptive analytics techniques were used to analyze how crime-related activities are changing across different regions and states. From the central tendency measures, it is clear that the mean values of apprehended cases, acquitted cases, and pending cases are greater than their median values. This implies that the distribution is positively skewed since the mean is greater than the median.

Additionally, using histograms, scatter plot, and box plot techniques, the study diagnosed how crime is being handled in different states or regions. From the analysis, it is clear that the relationship between apprehended cases and pending cases implies that states with more apprehended cases also experience more pending cases. Similarly, the study reveals that crime handling is not evenly distributed across different states. This implies that better case processing techniques and balanced law enforcement strategies should be implemented to reduce the backlogs experienced in different states.